1 Data Sources and Extraction

ProtEvidenceDB consolidates data from five biomedical APIs and public databases. The databases and the specific fields and query structures were chosen by our team; an LLM was used to generate the extraction scripts that execute these queries and parse the responses into CSV files.

1.1 EDRN (Early Detection Research Network)

Cancer biomarker records registered by the National Cancer Institute.

```
GET https://edrn.cancer.gov/data-and-resources/biomarkers/?ajax=json

Filters: kind in {Gene, Protein, Proteomic, Genomic, Genetic}
Fields: title, kind, organs, phases, description
```

Fields extracted: BiomarkerID (generated), Name, Type (mapped from `kind`), Organ, Phase (highest validation phase), Description.

1.2 UniProt (Universal Protein Resource)

Reviewed human protein records queried via the UniProt REST API.

```
GET https://rest.uniprot.org/uniprotkb/search
  ?query=(organism_id:9606) AND (gene_exact:"EGFR" OR ...)
  &fields=accession,id,gene_names,protein_name,
    organism_name,length,mass,cc_function,cc_disease,
    cc_subcellular_location,cc_tissue_specificity,
    ft_domain,ft_binding,ft_act_site,xref_alphafolddb,
    xref_pdb,xref_drugbank,xref_chembl,xref_kegg,
    xref_reactome,reviewed
  &format=json&size=500

Batched: 20 gene symbols per request
```

Fields extracted: UniProtID, UniProtAccession, EntryType, GeneSymbol, GeneSynonyms, ProteinName, Function, DiseaseAssociations, SubcellularLocation, TissueSpecificity, Domains, BindingSites, ActiveSites, AlphaFoldID, PDB_IDs, DrugBankIDs, ChEMBL_IDs, SequenceLength, MolecularWeight, Organism.

1.3 DGIdb (Drug Gene Interaction Database)

Drug–gene interactions queried via the DGIdb GraphQL API.

```
POST https://dgidb.org/api/graphql

query {
  genes(names: ["EGFR", "TP53", ...]) {
    nodes {
      name
      interactions {
        drug {
          name, conceptId, approved
          drugAttributes { name, value }
        }
        interactionScore
        interactionTypes { type, directionality }
        publications { pmid }
        sources { sourceDbName }
      }
    }
  }
}

Batched: 25 genes per request
```

Fields extracted:

- **Drugs:** DrugID (generated), DrugDBID (`conceptId`), Name, IsApproved.
- **Interactions:** InteractionID (generated), GeneSymbol, DrugID, InteractionType, Directionality, Score, Sources.
- **InteractionSources:** IS_ID (generated), InteractionID, PMID (one row per publication per interaction).

1.4 KEGG and Reactome (Biological Pathways)

Pathway data retrieved via two bulk KEGG REST API calls and UniProt Reactome cross-references.

```
# KEGG: all human gene-to-pathway links (single bulk request)
GET https://rest.kegg.jp/link/pathway/hsa

# KEGG: all human pathway names (single bulk request)
GET https://rest.kegg.jp/list/pathway/hsa

# Reactome pathways extracted from UniProt cross-references:
# xref_reactome field returns e.g.
# R-HSA-69620:Cell Cycle Checkpoints
```

Fields extracted: PathwayID (generated), Name, PathwayDBID (Reactome `R-HSA-*` or KEGG `hsa*`), Description.

1.5 OpenAlex (Journal Rankings)

Journal ranking metadata for oncology journals.

```
GET https://api.openalex.org/sources
  ?search=oncology&per_page=50&filter=type:journal

Repeated for 20 cancer-related search terms
  (e.g., "cancer research", "clinical oncology",
   "immunotherapy cancer", etc.)
```

Fields extracted: JournalID (generated), Name, ISSN, SCI_Ranking (`h_index`), ImpactFactor (`2yr_mean_citedness`), Publisher.

1.6 PubMed / NCBI E-Utilities

Oncology paper metadata fetched using tiered biomarker–drug queries. Each tier maintains the biomarker–drug pairing so that every paper can be traced back to a specific relationship for stance labeling. The two API endpoints used are:

```
# ESearch: find PMIDs matching a query
GET https://eutils.ncbi.nlm.nih.gov/entrez/eutils/esearch.fcgi
  ?db=pubmed&retmode=json&retmax=15&term=<QUERY>

# EFetch: retrieve full metadata for matched PMIDs
GET https://eutils.ncbi.nlm.nih.gov/entrez/eutils/efetch.fcgi
  ?db=pubmed&id=33590279,33123456,...&retmode=xml
```

The five query tiers, from most precise to broadest, are shown below. Each tier preserves which biomarker and drug the paper relates to.

```
Tier 1 (gene + drug, most precise):
  "EGFR"[Title/Abstract] AND "ERLOTINIB"[Title/Abstract]
```

```
Tier 2 (gene aliases + drug):
  ("EGFR" OR "ERBB1" OR "Epidermal growth factor receptor")
    [Title/Abstract] AND "ERLOTINIB"[Title/Abstract]
```

```
Tier 3 (gene + mechanism class + cancer):
  ("KRAS" OR "Kirsten ras oncogene")[Title/Abstract]
    AND "inhibitor"[Title/Abstract]
    AND cancer[Title/Abstract]
```

```
Tier 4 (organ + gene + drug):
  ("lung cancer" OR "NSCLC")[Title/Abstract]
    AND "EGFR"[Title/Abstract]
    AND "ERLOTINIB"[Title/Abstract]
```

```
Tier 5 (organ + gene + generic therapy, broadest):
  ("breast cancer")[Title/Abstract]
    AND ("BRCA1" OR synonyms)[Title/Abstract]
    AND ("drug" OR "therapy" OR "inhibitor")[Title/Abstract]
```

Fields extracted: PaperID (PMID), Title, DOI, PublicationDate, JournalName, JournalISSN, PublicationTypes, Abstract.

2 Entity Descriptions and Assumptions

This section describes each entity in the database. Following ER conventions taught in this course, entities represent a *person, place, thing, or event.* Relationships (Section 3) describe how entities interact or depend on each other. Foreign keys are omitted from entity descriptions as they are implied by the relationship lines in the ER diagram and belong to the implementation level, not the conceptual level.

2.1 Journal

A **Journal** represents a scientific publication venue (a *thing*) in which oncology papers appear. Multiple papers can be published in the same journal, and including journal-level attributes (ranking, impact factor, publisher) within each paper record would introduce significant redundancy. Separating Journal as its own entity ensures that journal metadata is stored once and referenced via the *publishes* relationship.

Assumptions:

- Each journal is uniquely identified by a generated JournalID; its ISSN serves as a secondary unique identifier.
- Journal rankings (SCI_Ranking, ImpactFactor) are static per extraction and may be updated in future cycles.
- A journal may have zero or more papers in the database; a paper belongs to exactly one journal.

2.2 Paper

A **Paper** represents a PubMed-indexed publication (a *thing*) relevant to cancer biomarker–drug research. Papers are the primary carriers of evidence in the system: each paper may describe findings about one or more biomarker–drug relationships.

Assumptions:

- Each paper is uniquely identified by its PubMed ID (PMID).
- Papers may have different study types (clinical trial, review, meta-analysis) recorded in PublicationType, which informs evidence quality assessment.
- A paper may be associated with multiple drug–gene interactions through the *evaluates* relationship.

2.3 Pathway

A **Pathway** represents a biological signaling or metabolic pathway (a *thing*) sourced from Reactome or KEGG. Multiple drugs may target the same pathway through different mechanisms, and multiple proteins or biomarkers may participate in the same pathway. Including pathway information as attributes within Drugs or Proteins would cause substantial redundancy, making Pathway a necessary independent entity.

Assumptions:

- Each pathway is uniquely identified by a generated PathwayID; its external database identifier is stored in PathwayDBID.
- A drug may be associated with many pathways, and a pathway may be associated with many drugs (M : M via the *targets* relationship).

2.4 Biomarker

A **Biomarker** represents a cancer-relevant biological indicator (a *thing*) registered in the EDRN database. Not all biomarkers are proteins, and not all proteins are biomarkers. EDRN biomarkers include genomic markers, proteomic markers, and multi-marker panels. A biomarker is related to proteins through the gene it represents; since multiple proteins may be associated with the same gene, a single biomarker could relate to multiple protein records. Including biomarker attributes within the Protein entity would require repeating the biomarker for each associated protein. Therefore, Biomarker is modeled as a separate entity linked to Proteins through the *maps_to* relationship.

Assumptions:

- Each biomarker is uniquely identified by a generated BiomarkerID.
- Type distinguishes between genomic and proteomic biomarkers.
- Phase represents the EDRN validation phase (1–5), indicating clinical maturity.
- A biomarker may be associated with zero or more proteins (via *maps_to*).

2.5 Protein

A **Protein** represents a reviewed human protein record (a *thing*) from UniProt. Given that multiple proteins may be targeted by the same drug, including protein-related information within the Drug entity would cause extensive redundancy, and vice versa. A single protein may interact with many drugs. Therefore, Protein is an independent entity.

Assumptions:

- Each protein is uniquely identified by its UniProt mnemonic ID (e.g., P53_HUMAN).
- UniProtAccession provides a stable secondary identifier for cross-referencing.
- Structural fields (Domains, BindingSites, ActiveSites, PDB_IDs) are stored as delimited strings; further normalization of these multi-valued attributes is deferred to future iterations.
- DrugBankIDs and ChEMBL_IDs serve as cross-reference identifiers to external drug databases.

2.6 Drug

A **Drug** represents a pharmaceutical compound (a *thing*) that interacts with one or more gene targets. Drug records are extracted from DGIdb and are kept minimal: the identifier, name, external database ID, and FDA approval status. Clinical trial phase and mechanism of action are not stored as static drug attributes; instead, they are derived at query time from the linked papers and interactions.

Assumptions:

- Each drug is uniquely identified by a generated DrugID.
- DrugDBID stores the external concept identifier (e.g., `chembl:CHEMBL553`).
- A drug may interact with many gene targets and may be associated with many pathways.

2.7 DrugGeneInteraction

A **DrugGeneInteraction** records a pharmacological interaction *event* between a drug and a specific protein target. The same broad-spectrum drug may react with different proteins with different interaction types (inhibitor, agonist, antibody, etc.) and directionalities. Embedding this information as attributes of either Drug or Protein would produce significant redundancy. An interaction is a real-world event with substantial independent attributes (type, directionality, confidence score, reporting sources), and it is itself referenced by other entities and relationships (InteractionSource, *evaluates*). This justifies its status as an entity rather than a simple relationship line.

Assumptions:

- Each interaction links exactly one Drug to one Protein.

- InteractionType and Directionality describe the pharmacological mechanism.
- Score is a DGIdb-computed confidence metric (higher indicates more corroborating sources).
- Sources lists the databases that report this interaction (e.g., ChEMBL, DrugBank, CIViC).

2.8 InteractionSource

An **InteractionSource** records a specific PubMed citation (a *thing*) that provides evidence for a given drug–gene interaction. A single interaction may be supported by multiple publications. Storing PMIDs as a delimited string within the interaction record would violate first normal form; therefore, each citation is extracted into a dedicated entity with its own meaningful attribute (PMID).

Assumptions:

- Each record links one InteractionID to one PMID.
- The PMID references a paper that may or may not exist in the Paper table (DGIdb PMIDs are not guaranteed to overlap with our PubMed extraction).

2.9 User

A **User** represents a registered account (a *person*) that can create annotations over existing records in the database. Users do not create or upload papers.

Assumptions:

- Each user has a unique UserID, a display name, and a unique email address.
- Three seed users are pre-populated for development purposes.

2.10 UserAnnotation

A **UserAnnotation** records a free-text note (a *thing*) attached by a user to any entity in the database (a protein, paper, or drug). This uses a polymorphic design: the AnnotationType field specifies which entity table the EntityID refers to. A UserAnnotation has substantial independent attributes (AnnotationType, EntityID, Annotation text) and spans multiple target entity types, justifying it as an entity rather than a simple relationship.

Assumptions:

- AnnotationType is constrained to: `Protein`, `Paper`, or `Drug`.
- EntityID references the primary key of the corresponding entity table.
- A user may create multiple annotations; each annotation belongs to exactly one user.

7 Data Scale Summary

Table	Expected Rows	Source
Biomarker	∼1,324	EDRN
Protein	∼1,113	UniProt
ProteinGeneBiomarkerRelation	∼1,119	Derived (maps_to)
Drug	∼6,786	DGIdb
DrugGeneInteraction	∼13,189	DGIdb
InteractionSource	∼5,000+	DGIdb PMIDs
Pathway	∼1,476	Reactome + KEGG
DrugPathway	∼285,902	Derived (targets)
Paper	∼8,000+	PubMed
Journal	∼343	OpenAlex
Stance	∼8,000+	Derived (evaluates)
User	3 (seed)	Generated
UserAnnotation	∼500	To be created during semester

The raw Biomarker, Drug, and Paper tables each exceed 1,000 rows, satisfying the course scale requirement. The DrugPathway junction table exceeds 285,000 rows due to the combinatorial nature of drug–gene–pathway relationships.

BioLitRecs

Search Evidence Annotate

CANCER BIOMARKER LITERATURE RETRIEVAL

Find evidence for biomarker–drug relationships

Search 48,763 curated PubMed papers across 1,324 cancer biomarkers and 6,786 drugs — with 8,056 linked biomarker–drug–paper stances awaiting annotation.

Search gene, drug, or cancer type… **Search**

By Biomarker | By Drug | By Cancer Type | By Pathway

Try `EGFR`, `Cetuximab`, or `Lung`

1,324	6,786	48,763	8,056
BIOMARKERS	DRUGS	PAPERS INDEXED	STANCE RECORDS

Browse by Cancer Type

View all →

Breast	Prostate	Lung	Ovary	Pancreas	Colon
478 biomarkers	427 biomarkers	254 biomarkers	232 biomarkers	72 biomarkers	29 biomarkers

High-Evidence Biomarker–Drug Pairs

Browse all pairs →

EGFR
Cetuximab
Breast · Lung · Ovary · Prostate

Supporting	—
Opposing	—
Unclear	100%

ACTC1
Dexamethasone
Prostate

Supporting	—
Opposing	—
Unclear	100%

AGT
Enalapril Maleate
Breast

Supporting	—
Opposing	—
Unclear	100%

BioLitRecs

Search Evidence Annotate

Home › Lung · Breast · Ovary · Prostate › EGFR › Cetuximab

Breast · Lung · Ovary · Prostate EDRN Phase 3

EGFR + Cetuximab

72 stance records linked for this biomarker–drug pair. All stances are currently awaiting annotation.

Export CSV | Export BibTeX | Annotate Papers

- Supporting 0%
- Opposing 0%
- Unclear 100%

All Papers (10 shown of 72) | Supporting (0) | Opposing (0) | Unclear (72)

Sort: Most Recent

Activating mutations in the epidermal growth factor receptor underlying responsiveness of non-small-cell lung cancer to gefitinib.

Unclear

The New England Journal of Medicine 2004-May-20 DOI: 10.1056/NEJMoa040938

Most patients with non-small-cell lung cancer have no response to the tyrosine kinase inhibitor gefitinib, which targets the epidermal growth factor receptor (EGFR). However, about 10 percent of patients have a rapid and often dramatic clinical response. The molecular mechanisms underlying sensitivity to gefitinib are unknown. We searched for mutations in the EGFR gene in primary tumors from patients with non-small-cell lung cancer who had a response to gefitinib and found somatic mutations in the EGFR kinase domain...

PMID 15118073 | NSCLC | Mutation Study

View on PubMed

EGFR mutations in lung cancer: correlation with clinical response to gefitinib therapy.

Unclear

Science (New York, N.Y.) 2004-Jun-04 DOI: 10.1126/science.1099314

Receptor tyrosine kinase genes were sequenced in non-small cell lung cancer (NSCLC) and matched normal tissue. Somatic mutations of the epidermal growth factor receptor gene EGFR were found in 15 of 58 unselected tumors from Japan and 1 of 61 from the United States. Treatment with the EGFR kinase inhibitor gefitinib (Iressa) causes tumor regression in some patients with NSCLC, more frequently in Japan than in the United States. Mutations were found in tumors from all 9 patients with gefitinib-sensitive NSCLC...

PMID 15118125 | NSCLC | Gefitinib

View on PubMed

Impact of common epidermal growth factor receptor and HER2 variants on receptor activity and inhibition by lapatinib.

Unclear

BIOMARKER INFO

GENE SYMBOL
EGFR

UNIPROT ACCESSION
P00533

BIOMARKER TYPE
Genomic

CANCER ORGANS
Breast · Lung · Ovary · Prostate

EDRN PHASE
Phase 3

PROTEIN FUNCTION
[To be pulled from uniprot_proteins.csv — Function field]

DESCRIPTION
[To be pulled from biomarkers.csv — Description field]

DRUG INFO

DRUG NAME
CETUXIMAB

MODALITY
Antineoplastic Agents

MECHANISM OF ACTION
[To be pulled from drugs.csv — MechanismOfAction field]

APPROVED
Yes